

1. Motivation

Magnetism has always been an interesting topic for me. Specifically how different mechanisms like electron spin, orbital occupancy, and exchange interactions come together to produce a macroscopic magnetic effect. Designing new magnetic materials typically requires Density Functional Theory (DFT) calculations, which are generally used for computationally predicting magnetic behavior in materials are computationally expensive and slow.

This project investigates whether the magnetic ordering type of an inorganic compound can be predicted directly from its chemical composition using machine learning, without any DFT calculations.

2. Data Source

Primary Dataset: Materials Project database, queried via the mp-api Python client with `has_props=magnetism`, returning all compounds with magnetism data available.

- Total compounds retrieved: 18415
- Sampled up to 5,000 per ordering type using stratified random sampling (`random_state=42`). AFM has only 3415 materials because that's all The Materials Project had available.
- Final dataset: 18,415 compounds x 25 features

Ordering	Count	Description
FM (Ferromagnetic)	5000	spins all point in the same direction -> net magnetic moment
FiM (Ferrimagnetic)	5000	spins point in opposite directions but with unequal magnitudes -> still a net moment
NM (Nonmagnetic)	5000	no spin ordering at all
AFM (Antiferromagnetic)	3415	spins point in opposite directions with equal magnitudes -> cancel out, no net moment

Table 1. Magnetic Ordering Types in the Dataset

Data Enrichment: Compositional features generated using `matminer` and `pymatgen`.

- `ElementProperty` featurizer with the `Magpie` preset applied to each compound's chemical formula

- 12 Magpie features selected covering electronegativity, atomic radius, unfilled electron shells, d-orbital occupancy, ground-state magnetic moment, and space group number

3. Data Analysis :

Before analyzing the data was prepared as follows:

- Dropped columns: material_id, formula, spacegroup, and total_magnetization were removed. The first two are identifiers with no predictive value, and total_magnetization is a direct leakage feature.
- Feature scaling: The continuous features were scaled using StandardScaler. The scaler was fitted exclusively on the training set and then applied to the validation and test sets.
- Splitting the data: The dataset was split 70% / 15% / 15% using stratified sampling on ordering to ensure all four magnetic ordering types are proportionally represented in each split.

3.1 EDA

- **Magnetization Distributions by Ordering Type**

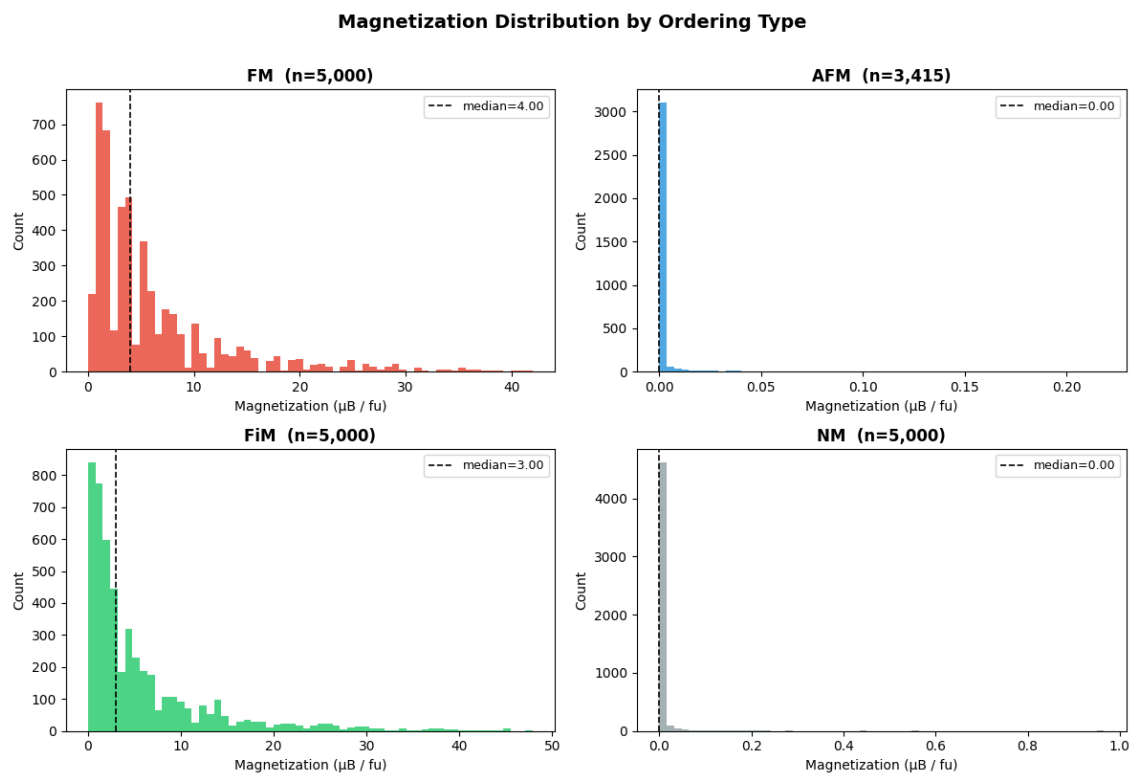


Figure 1. Magnetization Distribution by Ordering Type

Histograms of total magnetization normalized per formula unit (magnetization_per_fu) were plotted separately for each magnetization ordering type. Top 1% of values were trimmed to reduce the effect of outliers and median values were indicated on each plot. FM and FiM materials show clearly non-zero magnetization distributions, while NM materials are clustered around zero. AFM materials are also concentrated around zero, despite having magnetic ordering. This behavior is consistent with their spin-cancellation physics where equal and opposite spins result in no net magnetic moment.

- **Spearman Correlation Heatmap**

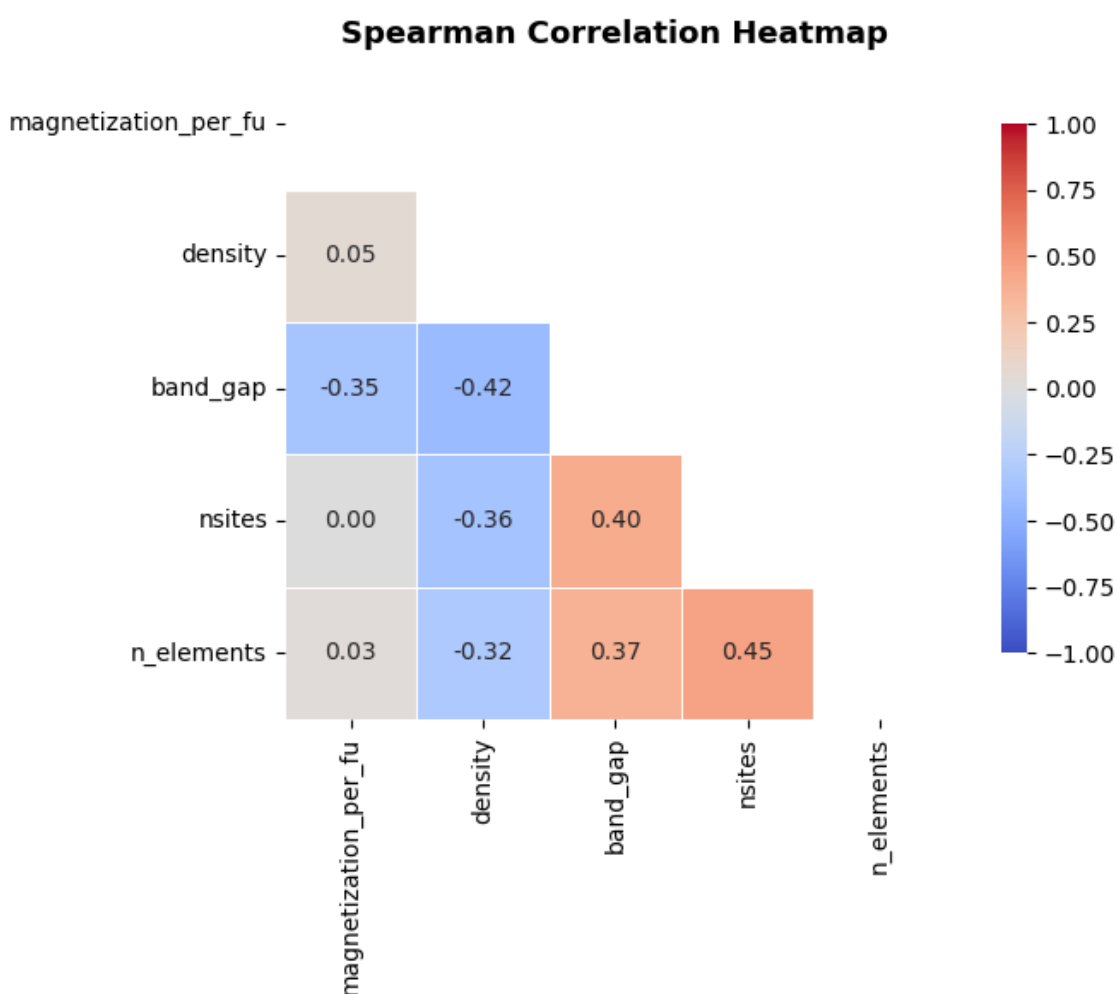


Figure 2. Spearman Correlation Heatmap

Spearman correlation matrix was computed across the base numerical features (magnetization_per_fu, density, band_gap, nsites, n_elements). Spearman was chosen over Pearson due to the magnetization distributions being heavily right-skewed. Only the lower triangle of the matrix is shown to avoid redundancy.

The heatmap reveals that `magnetization_per_fu` has weak correlations with all base features, with `band_gap` showing the strongest relationship ($\rho = -0.35$). Consistent with the expectation that magnetic materials tend to be metals with zero or near-zero band gap, `density` and `n_elements` show near-zero correlation with magnetization ($\rho = 0.05$ and 0.03 respectively), suggesting they carry little predictive signal on their own. The strongest correlation in the matrix is between `nsites` and `n_elements` ($\rho = 0.45$), which is expected as compounds with more sites tend to contain more distinct elements.

- **Magnetic Element Fraction**

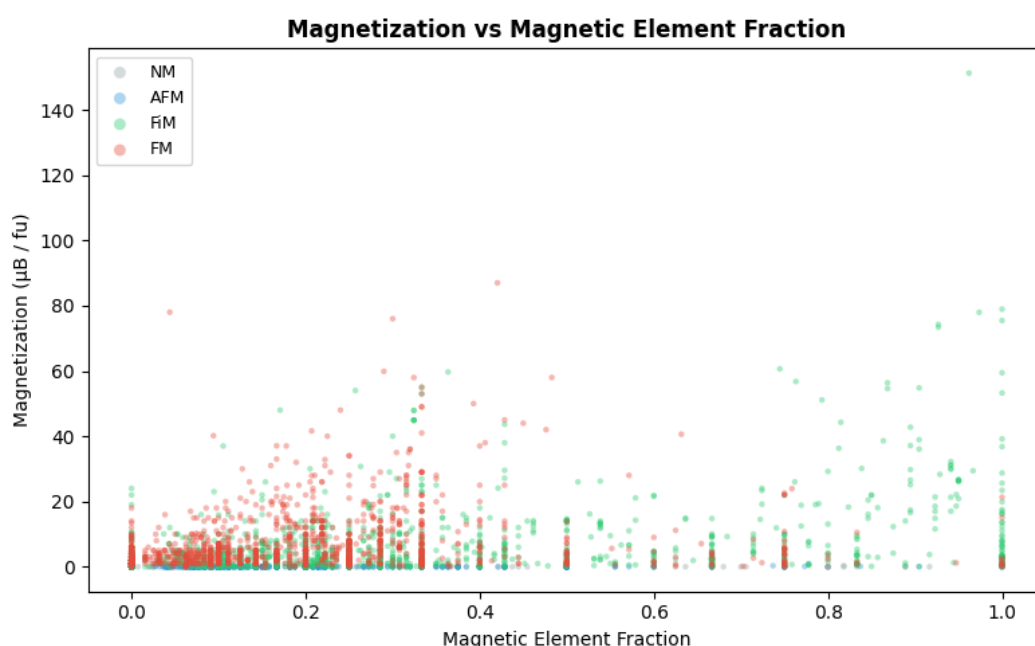


Figure 3. Magnetization vs. Magnetic Element Fraction

A custom feature was engineered representing the fraction of atoms in a compound belonging to known magnetic elements, 3d transition metals (Fe, Co, Ni, Mn, Cr, Cu, V) and rare earth elements (Gd, Tb, Dy, Ho, Er, Tm, Nd, Sm, Eu, Yb, Ce, Pr). A scatter plot of this feature against magnetization (5,000 randomly sampled points) shows that NM materials are densely clustered near zero magnetization across all fractions, while FM and FiM materials span a wider range of magnetization values and become more prevalent at higher magnetic element fractions. Notably, compounds with a magnetic element fraction approaching 1.0 are almost exclusively FM or FiM, confirming that a higher proportion of magnetic elements is strongly associated with non-zero magnetization. However, the large spread at intermediate fractions (0.2-0.6) indicates that magnetic element fraction alone is

not sufficient to predict ordering type, motivating the inclusion of additional compositional features.

• Magpie Feature Distributions

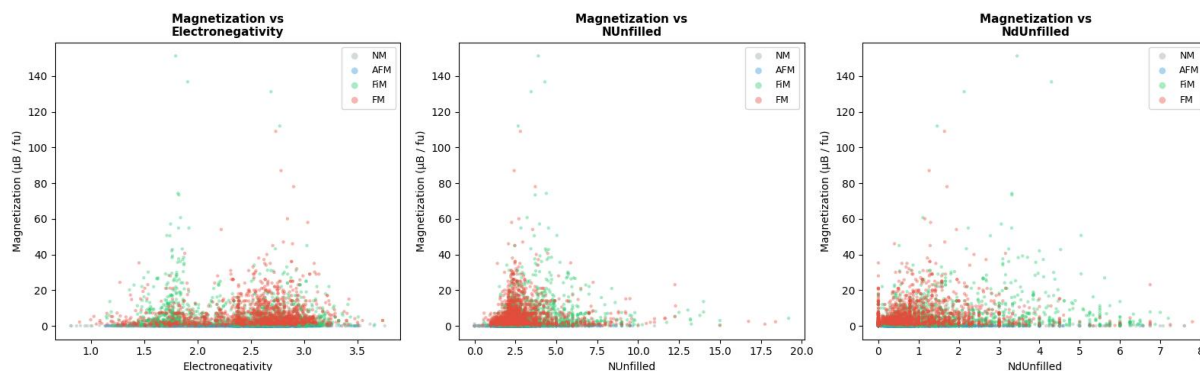


Figure 4. Feature Distributions

Three Magpie features were examined via scatter plots against magnetization (5,000 randomly sampled points, colored by ordering type):

- **Mean Electronegativity:** NM materials are concentrated in the 1.5-2.5 electronegativity range with near-zero magnetization, while FM and FiM materials are more spread out. There is no strong directional trend, suggesting electronegativity alone is a weak predictor of magnetization magnitude but may still help separate ordering types.
- **Mean NUnfilled:** Most compounds cluster at low NUnfilled values (0–5), with FM and FiM materials showing higher magnetization in this range. NM materials remain near zero across all NUnfilled values. The spread increases at higher NUnfilled values, indicating that more unfilled electron shells are associated with greater variance in magnetization.
- **Mean NdUnfilled:** This is the most discriminative of the three features. FM and FiM materials with non-zero magnetization are almost exclusively found at NdUnfilled values between 1-4, corresponding to partially filled 3d shells in transition metals like Fe, Co, and Ni. NM materials cluster at NdUnfilled ~ 0, consistent with the absence of unpaired d-electrons in non-magnetic compounds.

3.2 Hypothesis Testing

Four statistical tests were conducted to confirm that the engineered features are meaningfully related to magnetic ordering and magnetization. All tests use $\alpha = 0.05$ as the significance level.

H1: Does the average number of unfilled electron orbitals differ across ordering types?

Unfilled electron orbitals are a proxy for electrons available for magnetic exchange interactions. A Kruskal-Wallis test was used to compare the distribution of Mean NUnfilled across all four ordering types.

- H0: No significant difference in Mean NUnfilled across FM, AFM, FiM, and NM
- H1: At least one ordering type has a significantly different distribution
- Result: H0 rejected ($p = 2.08\text{e-}190 < 0.05$)

H2: Does the average number of unfilled d-orbitals differ across ordering types?

Transition metal magnetism arises from partially filled 3d shells, making Mean NdUnfilled a physically motivated feature. A Kruskal-Wallis test was applied across all four ordering types ($H = 2255.58$, $p = 0.00\text{e}+00$), followed by pairwise Mann-Whitney U tests to identify which groups differ. FiM materials have the highest median NdUnfilled (1.33), followed by FM (0.80) and AFM (0.73), while NM materials have a median of 0.0. All pairwise comparisons are significant except FM vs AFM ($p = 0.087$), both FM and AFM ordering are driven by similar transition metal d-electrons, differing in spin alignment rather than d-orbital occupancy.

- H0: No significant difference in Mean NdUnfilled across FM, AFM, FiM, and NM
- H1: At least one ordering type has a significantly different distribution
- Result: H0 rejected ($p=0.00\text{e}+00 < 0.05$)

H3: Does band gap differ between magnetic and non-magnetic materials?

Many magnetic materials are metals with a band gap of zero, while non-magnetic materials include insulators and semiconductors with non-zero band gaps. A Mann-Whitney U test was used to compare band gap distributions between magnetic (FM, AFM, FiM combined) and non-magnetic (NM) groups.

- H0: No significant difference in band gap between magnetic and non-magnetic materials
- H1: Band gap distributions differ between magnetic and non-magnetic materials
- Result: H0 rejected ($p = 5.58\text{e-}94 < 0.05$)

H4: Does magnetic element fraction correlate with magnetization?

Tests whether the engineered `mag_element_fraction` feature has a monotonic relationship with `magnetization_per_fu`. A Spearman correlation test was used as the relationship is not expected to be linear.

- H0: No monotonic relationship between magnetic element fraction and magnetization
- H1: A significant positive monotonic relationship exists
- Result: H0 rejected ($p = 0.348$, $p = 0.00e+00 < 0.05$)

All four null hypotheses were rejected, confirming that the engineered features carry statistically significant information about magnetic ordering and magnetization. These results directly motivate their inclusion in the ML classification model.

3.3 Machine Learning

Methodology

- Train/ Validation / Test split: 70% / 15% / 15% (12,890 / 2,762 / 2,763) with stratification on ordering type.
- StandardScaler fitted on training data only to prevent data leakage.
- Model selection based on validation accuracy.
- Random Forest used OOB scoring during training as an additional internal validation estimate.
- 5-fold cross-validation performed on the combined train+validation set to confirm generalizability.

Model Comparison

Four classifiers were trained and evaluated on the validation set. Then the models were compared to find the best classifier.

Model	Validation Accuracy
Logistic Regression	0.592
Gradient Boosting (125 trees)	0.696
XGBoost (200 trees)	0.703
Random Forest (300 trees)	0.723

Table 2. ML Model Validation Accuracies

Random Forest achieved the highest validation accuracy and was selected as the final model. The gap between Logistic Regression and the tree-based models suggests that the decision

boundaries between ordering types are non-linear and cannot be captured by a linear classifier alone.

Random Forest Test Set Results

Class	Precision	Recall
AFM	0.69	0.65
FM	0.57	0.60
FiM	0.67	0.66
NM	0.86	0.85

Table 3. Random Forest Results

Performance varies considerably across classes. NM achieves the highest precision and recall (0.86 / 0.85), while FM is the weakest (0.57 / 0.60), with AFM and FiM falling in between. The model is both accurate and thorough in identifying NM compounds, but struggles most with FM, it both mislabels other compounds as FM and misses actual FM compounds. The slight drop from validation accuracy (0.723) to test accuracy (0.694) is moderate, indicating no significant overfitting.

Random Forest 5-Fold Cross Validation

Fold	Accuracy
Fold 1	0.7126
Fold 2	0.7065
Fold 3	0.7054
Fold 4	0.7051
Fold 5	0.7208

Table 4. 5-Fold Cross Validation Accuracies

The overall mean for the 5-Fold cross validation is 0.7101, with the standard deviation of ± 0.0060 . The low standard deviation across folds confirms that results are stable and the model generalizes reliably to unseen data.

4. Findings

Hypothesis Testing

All four null hypotheses were rejected with very small p-values, confirming that the engineered features carry statistically significant information about magnetic ordering and magnetization.

Mean NUnfilled ($p = 2.08\text{e-}190$) and Mean NdUnfilled ($p \sim 0$) differ significantly across ordering types, with the notable exception that FM and AFM are not significantly different from each other in NdUnfilled ($p = 0.087$), consistent with both being driven by similar d-electron configurations and differing only in spin alignment. Band gap differs significantly between magnetic and non-magnetic materials ($p = 5.58\text{e-}94$), and magnetic element fraction shows a significant positive monotonic correlation with magnetization ($p = 0.348$, $p \sim 0$).

Classification Performance

Magnetic ordering type can be predicted from compositional features alone with 72.3% validation accuracy and 69.4% test accuracy using a Random Forest classifier. Results are stable across 5-fold cross-validation (mean 0.710 ± 0.006), confirming reliable generalization to unseen data.

Per-Class Performance

- NM is the easiest class to predict (precision 0.86, recall 0.85), non-magnetic materials are clearly separable through near-zero band gap and low magnetic element fraction, making them compositionally distinct from the other three types.
- AFM performs moderately (precision 0.69, recall 0.65). Antiferromagnetic compounds share similar transition metal compositions with FM and FiM, making them harder to distinguish.
- FiM performs moderately (precision 0.67, recall 0.66). Frequently confused with FM as both ordering types are dominated by the same magnetic transition metals.
- FM is the hardest class to predict (precision 0.57, recall 0.60). FM and FiM share nearly identical compositional signatures. The difference between them is spin alignment, which is a structural and quantum mechanical property that cannot be inferred from chemical composition alone.

Confusion Matrix

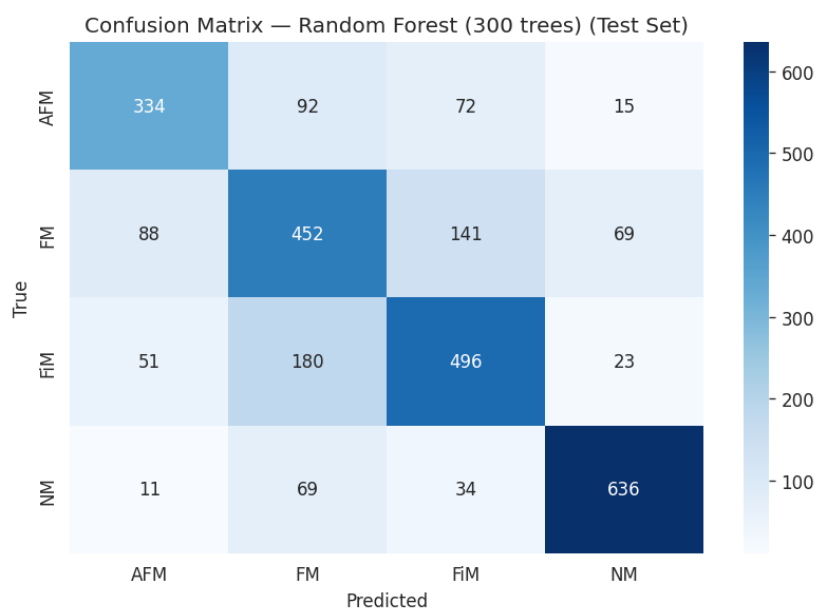


Figure 5. Random Forest Confusion Matrix

The confusion matrix on the test set confirms the per-class patterns. NM is the most accurately predicted class with 636 out of 750 correct, with only 114 misclassifications spread across the other classes. FM and FiM show the most confusion with each other: 141 FM compounds are misclassified as FiM and 180 FiM compounds are misclassified as FM, reflecting their near-identical compositional signatures. AFM performs moderately with 334 correct out of 513 but shows notable confusion with both FM (92) and FiM (72).

Feature Importance

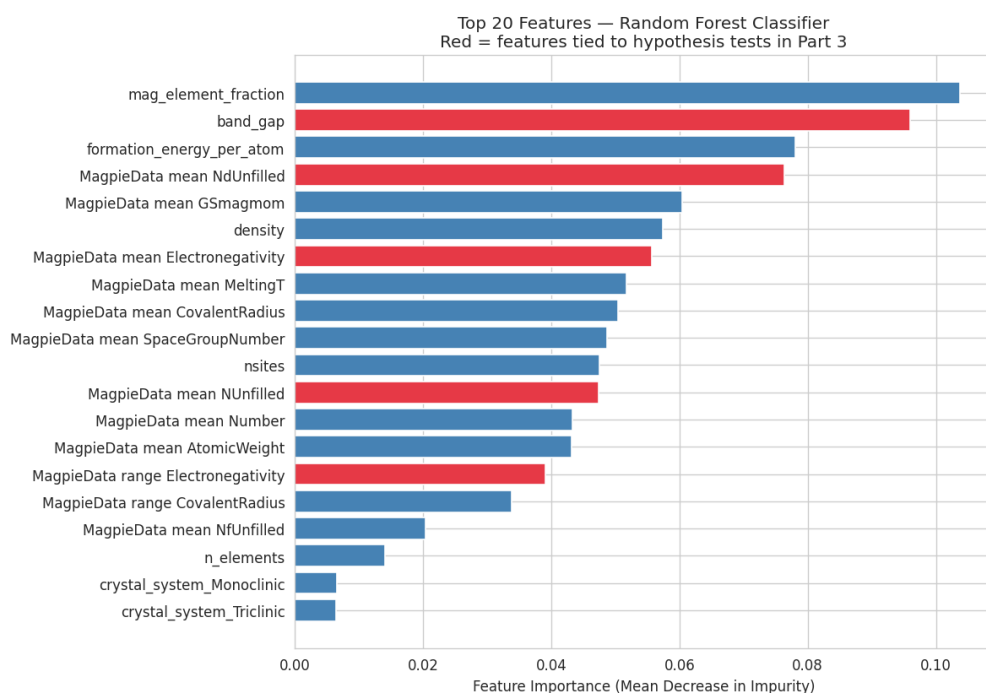


Figure 6. The top 20 features ranked by mean decrease in impurity (Red bars indicate features linked to hypothesis tests.)

`mag_element_fraction` (H4) and `band_gap` (H3) are the two strongest predictors with importances of 0.104 and 0.096 respectively. `MagpieData mean NdUnfilled` (H2) ranks 4th (0.076), while `MagpieData mean NUnfilled` (H1) ranks 12th (0.047). All four hypothesis-tested features appear in the top 12, showing a direct correspondence between the statistical significance found in hypothesis testing and prediction of the classification model.

5. Limitations & Future Work

The primary limitation of this project is that all features are purely compositional, they describe the types and proportions of elements in a compound but carry no information about atomic positions, bond angles, or structural relaxation. This is the main reason FM and FiM are difficult to distinguish: both are dominated by the same magnetic transition metals, and their difference is a quantum mechanical property of spin alignment that requires structural context to capture.

Magpie features do not capture bonding geometry or local coordination environments, which are known to influence magnetic behavior.

The AFM class is underrepresented (3,415 samples vs 5,000 for other classes) due to the limited availability of AFM compounds in the Materials Project database, which may affect model performance on this class.

Future work could explore using crystal structure directly as input rather than compositional summaries, adding more physics-based features, predicting other magnetic properties such as Curie temperature, and expanding the dataset as the Materials Project grows.